

Overlay — EVPN/VXLAN, VRFs, and External Access

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Status: Draft

Scope: BGP EVPN control plane, VXLAN data plane, day-1 VRFs (**storage**, **edge**), API ingress path, border handoff

Implements: [SPEC.md §5.1](#) (overlay row), [§6.3](#) (storage isolation), [§8](#) (request flow entry) · builds on [docs/network.md §5](#) underlay · consumed by [docs/serving.md §5](#)

SoT rule: all overlay addressing (VTEP loopbacks, VRF subnets, VNIs) lives in the seed IPAM (`bootstrap/netbox/seed/site.yaml`) and is rendered into switch configs by `bootstrap/seed/ztp/render.py` — same pipeline as the underlay; nothing is hand-configured.

1 Why an overlay, and why now

The day-1 fabric is a pure L3 eBGP underlay (network.md §5) — correct for RoCE rails, which stay **exactly as they are**. The overlay exists for the two things the underlay deliberately did not solve:

Need	Underlay answer	Overlay answer
Storage isolation (SPEC §6.3 “dedicated storage VRF”)	shared default VRF, DSCP only	VRF storage — storage nodes and worker mounts in their own routing table
External API access for ~1,000 users (SPEC §8)	none (no border)	VRF edge — LoadBalancer VIPs + campus border, no route to underlay/RoCE
Future re-search tenant VRFs, DCI	—	add a VRF + VNI, zero underlay change

What the overlay is *not* for: the RoCE expert-parallel path. GPU↔GPU NCCL/EP traffic stays native in the underlay rail subnets — no VXLAN encap, no PFC interaction changes, nothing from network.md §6 moves.

2 Control and data plane design

Element	Choice
Control plane	BGP EVPN over eBGP: leaf Lo0 ↔ spine Lo0, ebgp-multihop 3, send-community extended ; spines next-hop-unchanged (pure transit for VTEP next-hops)
Data plane	VXLAN, UDP 4789, VTEP = Loopback1 on every rail leaf + both spines
IRB model	Symmetric IRB — one L3VNI per VRF, routed type-5 everywhere; L2VNIs only for the two access VLANs
Underlay support	Lo1 advertised in ipv4 AF (already rendered); fabric MTU 9216 absorbs the 50 B VXLAN header at host MTU 9000
EOS mode	service routing protocols model multi-agent on all fabric switches (required for EVPN; part of rendered config, applied at ZTP)
Anycast gateway	VARP: shared .1 + per-leaf real IP (same pattern as the OOB pair) — real IPs also terminate MetalLB BGP (§5)

The existing p2p eBGP sessions keep doing the underlay (ipv4 + BFD). EVPN runs on separate loopback sessions: 2 per leaf, 8 per spine — 16 sessions total.

2.1 VNI / RD / RT plan

Object	VNI	RD	RT	Where
VRF storage (L3)	50200	<Lo0>:200	50200:50200	all 8 leaves
VRF edge (L3)	50050	<Lo0>:50	50050:50050	all 8 leaves + both spines
VLAN 200 storage-access (L2)	10200	<Lo0>:10200	10200:10200	all 8 leaves
VLAN 50 edge-access (L2)	10050	<Lo0>:10050	10050:10050	all 8 leaves

Convention: L2VNI = 10000 + VLAN ID, L3VNI = 50000 + VLAN ID of the primary access VLAN. Future tenant VRFs allocate the next VLAN + both VNIs from the same rule.

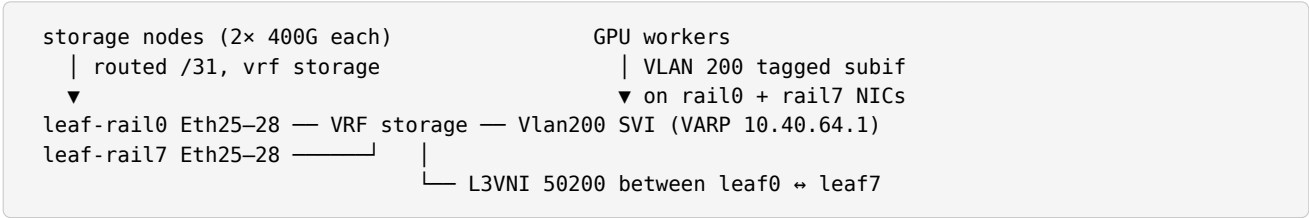
2.2 Addressing (seed IPAM)

Block	Use
10.30.32.0/24	VTEP loopbacks (Lo1): spines .1–.2, leaf-rail0–7 .11–.18
10.40.0.0/24	Storage p2p /31s, vrf storage : leaf-rail0 Eth25–28 = .0–.7, leaf-rail7 Eth25–28 = .8–.15
10.40.64.0/24	VLAN 200 storage access: VARP gw .1, leaf-rail <i>i</i> .2+ <i>i</i> , worker0 <i>N</i> .1 <i>N</i>

Block	Use
10.50.0.0/24	LoadBalancer VIP pool (MetalLB): inference gateway VIP .10 = inference.ai.local
10.50.64.0/24	VLAN 50 edge access: VARP gw .1, leaf- rail <i>i</i> .2+i, worker0 <i>N</i> .1 <i>N</i>
10.50.255.0/29	Border p2p /31s, vrf edge : spine1 Eth33 .0/31, spine2 Eth33 .2/31

3 VRF storage

Moves the network.md §7 storage attachment into its own routing table — same leaves, same ports, same QoS (DSCP 18 → TC1, lossy).



Item	Design
Storage node side	unchanged /31 routed ports on leaf-rail0/7 Eth25–28, now vrf storage
Worker side	VLAN 200 tagged subinterface on the rail0 and rail7 NICs (2 paths, ECMP/failover); IP 10.40.64.1N , gw VARP .1 — configured by cloud-init from NetBox (bootstrap track)
QoS	storage traffic still marked DSCP 18 on the subif → TC1; RoCE class untouched (untagged rail traffic keeps DSCP 26)
Route scope	10.40.0.0/16 exists only in vrf storage; workers reach it via the VLAN 200 subif, nothing in the default VRF routes to storage
Blast radius	a storage-side routing mistake can no longer leak into the RoCE underlay

Trunking note: rail leaf host ports are now `trunk native vlan <rail>` with tagged 50/200 allowed. Untagged RoCE frames land in the rail VLAN exactly as before; rails 1–6 simply carry no tagged traffic day-1.

4 VRF edge — API ingress and border

The path SPEC §8 needs: external users → API gateway VIP, with the fabric underlay invisible to them.

```

campus / users                                AS 65500 (firewall)
  | default route + 10.50.0.0/24 learned
  ▼
spine1/2 Eth33 (routed /31, vrf edge, eBGP)    ← border VTEPs
  | EVPN type-5
  ▼
leaf-rail0/7 · vrf edge · Vlan50 (VARP 10.50.64.1)
  | VLAN 50 tagged subif on worker rail0/7 NICs
  ▼
MetalLB speakers (AS 65200) — announce 10.50.0.0/24 VIPs
  ▼
Envoy gateway Service VIP 10.50.0.10 → InferencePool → vLLM pods

```

Item	Design
Border	spine Eth33 routed /31 in vrf edge; eBGP to campus firewall AS 65500 (default in, VIP /24 out)
Spines as VTEPs	yes — border VRF terminates on spines (Lo1 .1/.2); leaves reach it via L3VNI 50050
VIP announcement	MetalLB BGP mode (AS 65200) from workers over the VLAN 50 subif; leaves <code>bgp listen range 10.50.64.0/24</code> in vrf edge (rails 0/7)
Isolation	vrf edge contains only 10.50.0.0/16 + firewall default; no import of underlay, storage, mgmt, or OOB routes
In-cluster hop	gateway pods (hostNetwork=false) receive VIP traffic via kube-proxy from the MetalLB-announced node; East-west to vLLM pods rides the CNI on rail subnets as usual
DNS	<code>inference.ai.local</code> → 10.50.0.10 (seed dnsmasq + campus conditional forward)

Security stance: the fabric exposes exactly one /24 of VIPs to the outside; authentication, keys, and quotas happen at the Envoy gateway (serving.md §6) — the fabric’s job ends at delivering packets to the VIP in an isolated table.

5 Rendered config summary (what `render.py` now emits)

Switch	Overlay additions
all leaves	multi-agent model · Lo1 VTEP · VLANs 50/200 + VARP SVIs in vrf edge/storage · Vxlan1 (2× L2VNI + 2× L3VNI) · EVPN peer group → spine Lo0s · MAC-VRF + VRF stanzas in BGP
leaf-rail0/7 only	storage /31 ports Eth25–28 in vrf storage · MetalLB <code>bgp listen range</code> in vrf edge
spines	multi-agent model · Lo1 VTEP · vrf edge + Vxlan1 (L3VNI 50050) · EVPN peer group → all leaf Lo0s with <code>next-hop-unchanged</code> · border Eth33 eBGP AS 65500
OOB pair	no change — overlay never touches 7010TX-48

As with the RoCE profile, exact EVPN/VXLAN syntax is validated against the target EOS release in the lab before production (`network.md` §8 gate applies).

6 Verification gates

Gate	Command	Pass
EVPN sessions	<code>show bgp evpn summary</code>	leaves: 2 Established; spines: 8
VTEP discovery	<code>show vxlan vtep</code>	every leaf sees 7 leaf + 2 spine VTEPs
Type-5 routes	<code>show bgp evpn route-type ip-prefix 10.50.0.0/24</code>	present on spines with leaf VTEP next-hop
Storage VRF	<code>ping vrf storage 10.40.0.1 source 10.40.64.2 (leaf0→stor01)</code>	ok; same ping from default VRF fails
Anycast gw	worker: <code>ping 10.40.64.1</code> and <code>10.50.64.1</code> from subifs	ok via either rail0/7 path
Isolation	from campus: <code>traceroute to 10.30.64.11</code> (rail IP)	blocked (no route in vrf edge)
VIP path	<code>curl https://inference.ai.local/v1/models</code> from campus	200 via border → spine → leaf → MetalLB node
RoCE regression	re-run network.md §8.3 rail gates after overlay deploy	unchanged (encap never touches rail traffic)

7 Phase-next

Item	Trigger
Research tenant VRFs	per-team isolation ask; allocate VLAN/VNI per §2.1 convention
DCI	second site; extend EVPN via border (type-5 only)
Storage VRF on STOR-2	second storage rack (SPEC §2A.6)
Anycast VTEP / MLAG leaves	only if host dual-homing model changes — not planned

8 References

Doc / path	Use
docs/network.md §5-6	Underlay + RoCE profile the overlay rides on
docs/serving.md	What lives behind the edge VIP
bootstrap/seed/ztp/render.py	Rendered overlay config (SoT pipeline)
bootstrap/netbox/seed/site.yaml	VNI/VRF addressing (IPAM SoT)
bootstrap/platform/apps/api/	MetallB pool + gateway consuming vrf edge
Arista EOS <i>EVPN/VXLAN</i> chapters (target release)	Syntax validation gate

9 Revision History

Version	Date	Notes
0.1	2026-07-19	Initial: EVPN/VXLAN overlay, vrf storage + vrf edge, border + VIP ingress, rendered via ZTP pipeline